DO NOW — RECALL

Lesson 104 · Magnetic, Optical and Flash Storage

Name: _____ Class: _____ Date: _____

1. What does non-volatile mean?

2. What is seek time?

3. What does optical storage use to read data?

4. 1 byte contains how many bits?

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Retains contents without power.

2. Model answer

Time to position/locate data for a storage access.

3. Model answer

Light/laser.

4. Model answer

8 bits.

By the end, you can…

• Magnetic hard disks use magnetised regions on spinning platters with moving heads; they can offer large capacity at relatively low cost per GB.

• Optical discs are read with a laser, often used for physically distributed or archived/removable media, though capacity and drive availability vary.

• Flash uses electronic non-volatile memory cells without moving read/write parts; often faster random access and more robust to vibration than HDDs, but trade-offs remain.

Vocabulary — know these terms

magnetic — Storage via magnetisation.

optical — Storage read with laser/light.

flash — Non-volatile electronic solid-state storage.

cost per GB — Price relative to capacity.

NEW LEARNING

Read, annotate and explain

Understand

Magnetic hard disks use magnetised regions on spinning platters with moving heads; they can offer large capacity at relatively low cost per GB.

Apply

Optical discs are read with a laser, often used for physically distributed or archived/removable media, though capacity and drive availability vary.

Evaluate

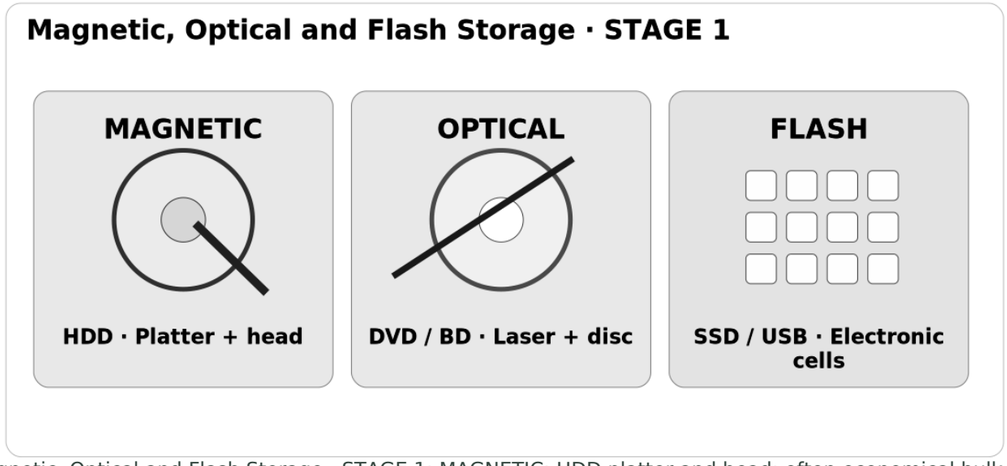
Flash uses electronic non-volatile memory cells without moving read/write parts; often faster random access and more robust to vibration than HDDs, but trade-offs remain.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

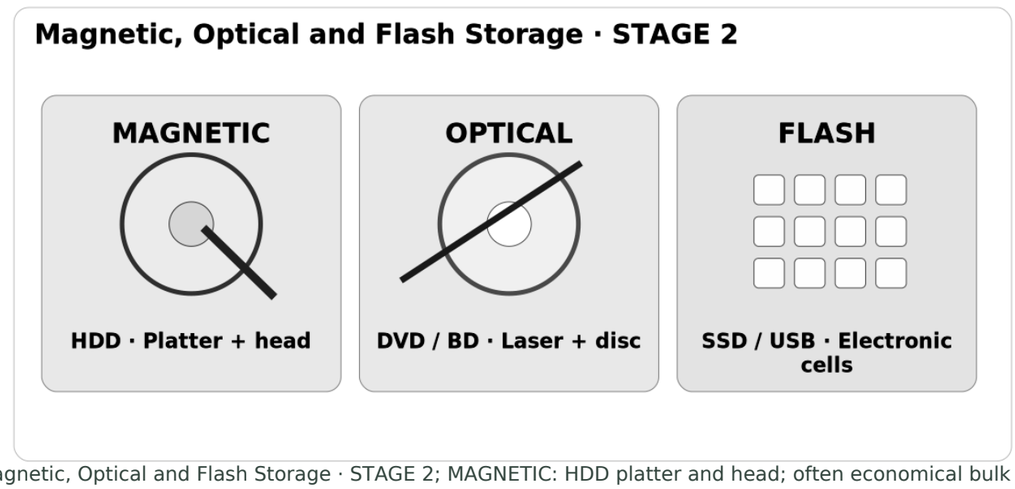
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Magnetic, Optical and Flash Storage · STAGE 1; MAGNETIC: HDD platter and head; often economical bulk storage

New Learning



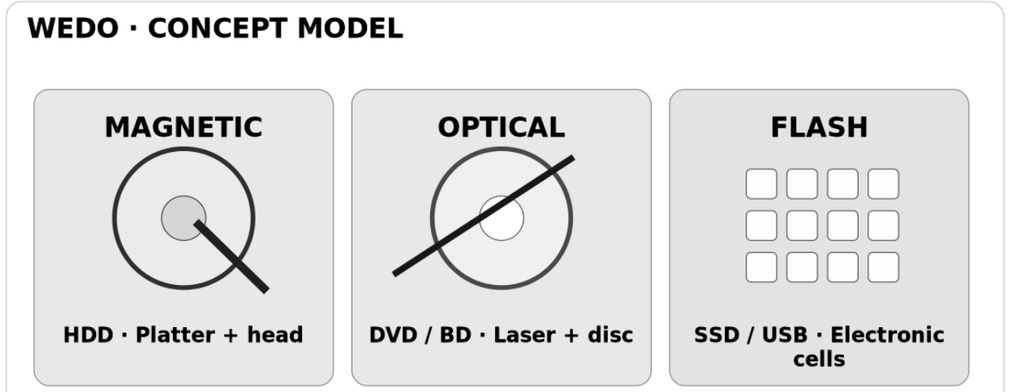
Magnetic, Optical and Flash Storage · STAGE 2; MAGNETIC: HDD platter and head; often economical bulk storage; OPTICAL: Disc read via laser; useful for removable media

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

A school wants inexpensive bulk storage for a large local video archive, with separate backups.

Magnetic HDD is a reasonable candidate because of capacity and cost per GB; validate resilience and transfer needs.



WEDO · CONCEPT MODEL; A school wants inexpensive bulk storage for a large local video archive, with separate backups.: Magnetic HDD is a reasonable candidate because of capacity and cost per GB; validate resilience and transfer needs.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

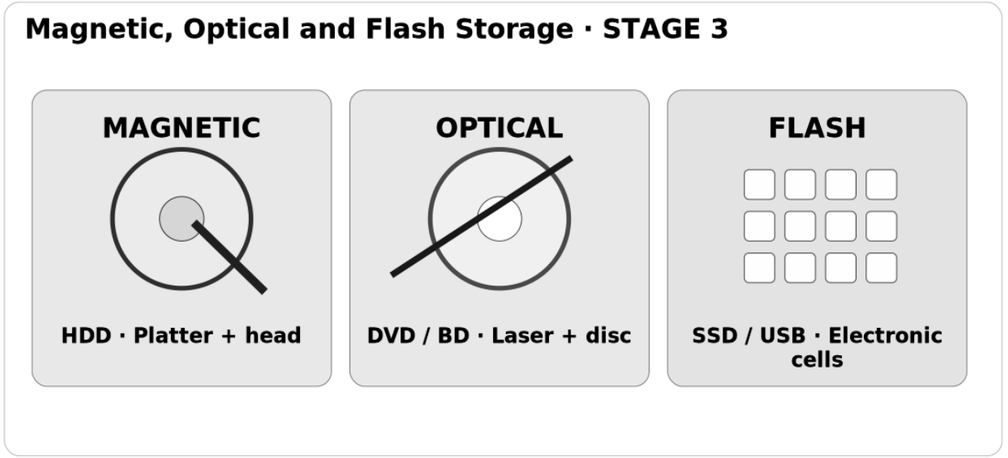
Guided 1 — CHECK

Explain the choice
Magnetic HDD is a reasonable candidate because of capacity and cost per GB; validate resilience and transfer needs.

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Magnetic, Optical and Flash Storage · STAGE 3; MAGNETIC: HDD platter and head; often economical bulk storage; OPTICAL: Disc read via laser; useful for removable media; FLASH: Solid-state cells; no moving read/write head

I DO — WORKED EXAMPLE

Follow the worked method

Choose portable storage for a field photographer travelling over rough ground with active projects.

Explain the choice

Flash SSD: no spinning platter and generally quick random access; still requires backups.

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

A hard disk drive reads bits using a laser.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

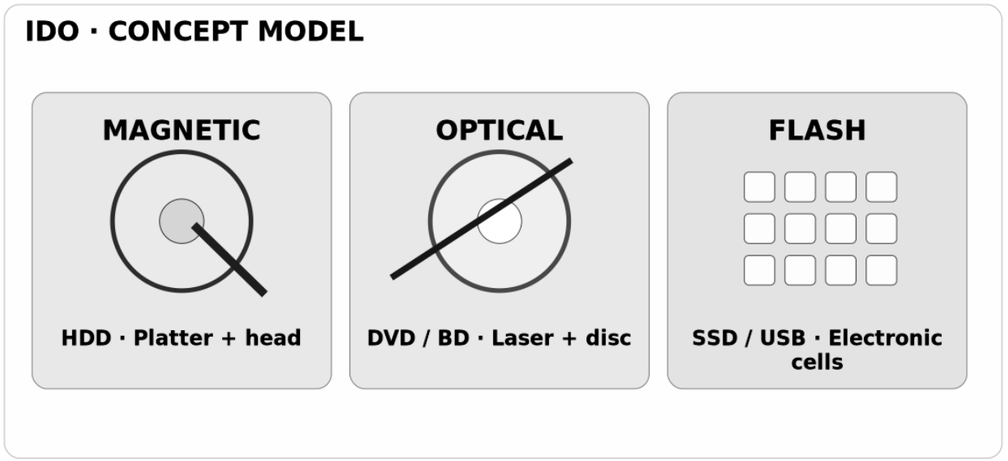
Evaluate storage choice for distributing 100 offline revision packs to pupils and for a multi-terabyte editable school video archive. [8]

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Choose portable storage for a field photographer travelling over rough ground with active projects.

Flash SSD: no spinning platter and generally quick random access; still requires backups.



IDO · CONCEPT MODEL; Choose portable storage for a field photographer travelling over rough ground with active projects.: Flash SSD: no spinning platter and generally quick random access; still requires backups.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

A school wants inexpensive bulk storage for a large local video archive, with separate backups.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Match HDD, DVD and USB SSD to magnetic, optical and flash storage, with mechanisms. [6]

Question 2

Explain one advantage and one limitation of flash SSD over magnetic HDD for a mobile video editor. [6]